



A Distributed Streaming Architecture for Global Music Mood Analysis

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1. Project Overview

This project proposes establishing a real-time data analytics pipeline to map time-dependent mood fluctuations based on music consumption habits. By combining user listening logs with the acoustic features of songs, the system will simulate a high-volume data stream via Apache Kafka. Subsequently, Apache Spark Streaming will process these continuous logs within sliding time windows to compute temporal changes in features such as songs' 'valence' (musical positivity) and 'energy'. The ultimate goal is to build an end-to-end distributed data engineering architecture that processes unstructured streaming data and visualizes how collective music preferences evolve at different times of the day on a live dashboard.

2. Source

<https://www.kaggle.com/datasets/a9santoshbhor/spotify-dataset>

<https://www.kaggle.com/datasets/sgoutami/spotify-streaming-history>

3. Dataset

Spotify Streaming History:

- **spotify_track_uri:** A unique identifier for each track on Spotify.
- **track_name:** The name of the track.
- **artist_name:** The name of the artist who performed the track.
- **album_name:** The name of the album the track belongs to.
- **ts:** Timestamp indicating when the playback of the track ended.
- **platform:** The device or platform used to stream the track (e.g., mobile, desktop, web player).
- **ms_played:** The duration of the track played in milliseconds.
- **reason_start:** The reason why the track started playing (e.g., user selection, autoplay, recommendation).
- **reason_end:** The reason why the track stopped playing (e.g., natural end, user skip, interruption).
- **shuffle:** Indicates whether shuffle mode was active during playback (TRUE or FALSE).
- **skipped:** Indicates whether the user manually skipped to the next track (TRUE or FALSE).

Spotify Dataset

- **valence:** A measure from 0.0 to 1.0 describing the musical positiveness conveyed by a track.
- **year:** The release year of the track.
- **acousticness:** A confidence measure from 0.0 to 1.0 of whether the track is acoustic.
- **artists:** The name of the artist or artists who performed the track.
- **danceability:** How suitable a track is for dancing based on a combination of musical elements including tempo and rhythm.
- **duration_ms:** The duration of the track in milliseconds.
- **energy:** A measure from 0.0 to 1.0 representing a perceptual measure of intensity and activity.
- **explicit:** Indicates whether the track contains explicit lyrics or adult content.
- **id:** A unique identifier for each track on Spotify.
- **instrumentalness:** Predicts whether a track contains no vocals, with values closer to 1.0 representing purely instrumental tracks.

4. Platforms, frameworks, tools, cloud systems, hardware

- **Message Broker / Data Ingestion:** Apache Kafka (To simulate a real-time, continuous streaming of listening logs and to buffer the data ingestion workload).
- **Distributed Processing Engine:** Apache Spark (Specifically PySpark and Spark Streaming API for sliding window calculations, real-time data joining operations, and distributed workload management).
- **Database:** PostgreSQL (To store processed, time-series-grouped, and aggregated metrics).
- **Visualization:** Streamlit, Grafana or Superset (To build a live "Mood Dashboard" by dynamically querying the database).
- **Containerization:** Docker (To package all services as independent and isolated images).
- **Orchestrating and Elastic Scaling:** Kubernetes (Minikube / kind) (To establish a local distributed cluster architecture, manage components, and enable auto-scaling of Spark worker pods based on load conditions during instant data flood simulations using Horizontal Pod Autoscaler [HPA]).

- **Hardware:** High-performance local workstations (The project will be executed within an isolated Kubernetes cluster run on local hardware resources, fully complying with the course's local execution constraints, and the system's resource limits will be tested).